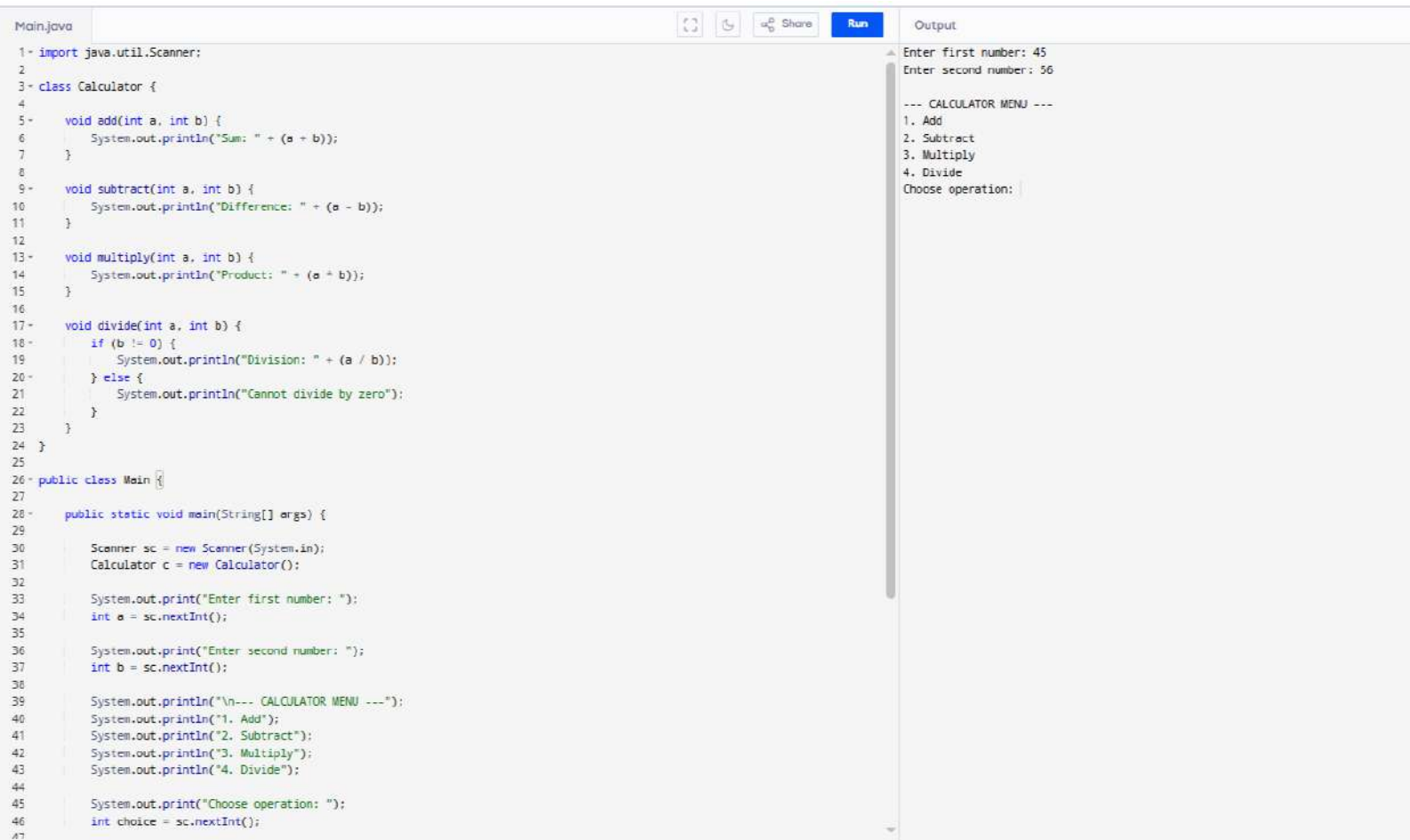


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Q:1
WRITE A JAVA PROGRAM TO CREATE A SIMPLE
CALCULATOR.



The screenshot displays a Java IDE with a file named 'Main.java'. The code defines a 'Calculator' class with methods for addition, subtraction, multiplication, and division, and a 'Main' class that uses a 'Scanner' to take user input and call these methods. The output window on the right shows the program's execution, including prompts for numbers, a menu of operations, and a prompt for the user's choice.

```
1- import java.util.Scanner;
2
3- class Calculator {
4
5-     void add(int a, int b) {
6         System.out.println("Sum: " + (a + b));
7     }
8
9-     void subtract(int a, int b) {
10        System.out.println("Difference: " + (a - b));
11    }
12
13-     void multiply(int a, int b) {
14        System.out.println("Product: " + (a * b));
15    }
16
17-     void divide(int a, int b) {
18        if (b != 0) {
19            System.out.println("Division: " + (a / b));
20        } else {
21            System.out.println("Cannot divide by zero");
22        }
23    }
24 }
25
26- public class Main {
27
28-     public static void main(String[] args) {
29
30         Scanner sc = new Scanner(System.in);
31         Calculator c = new Calculator();
32
33         System.out.print("Enter first number: ");
34         int a = sc.nextInt();
35
36         System.out.print("Enter second number: ");
37         int b = sc.nextInt();
38
39         System.out.println("\n--- CALCULATOR MENU ---");
40         System.out.println("1. Add");
41         System.out.println("2. Subtract");
42         System.out.println("3. Multiply");
43         System.out.println("4. Divide");
44
45         System.out.print("Choose operation: ");
46         int choice = sc.nextInt();
47     }
48 }
```

Output

```
Enter first number: 45
Enter second number: 56

--- CALCULATOR MENU ---
1. Add
2. Subtract
3. Multiply
4. Divide
Choose operation: 
```

Question:02

MAKE ANIMAL MAIN CLASS AND CATS AND DOGS CLASS AND CALL CATS AND DOGS FEATURES IN MAIN ANIMAL CLASS

```
Main.java  [Icons]  Share  Run  Output

1- class Animal {
2
3-   void eat() {
4       System.out.println("Animal is eating");
5   }
6
7-   void sleep() {
8       System.out.println("Animal is sleeping");
9   }
10 }
11
12 // ----- CAT CLASS -----
13 class Cat extends Animal {
14
15-   void meow() {
16       System.out.println("Cat says: Meow Meow");
17   }
18 }
19
20 // ----- DOG CLASS -----
21 class Dog extends Animal {
22
23-   void bark() {
24       System.out.println("Dog says: Woof Woof");
25   }
26 }
27
28 // ----- MAIN CLASS -----
29 public class Main {
30
31-   public static void main(String[] args) {
32
33       Animal a;
34
35       System.out.println("----- CAT -----");
36       a = new Cat();
37       a.eat();
38       a.sleep();
39       ((Cat) a).meow(); // calling Cat feature
40
41       System.out.println("\n----- DOG -----");
42       a = new Dog();
43       a.eat();
44       a.sleep();
45       ((Dog) a).bark(); // calling Dog feature
46   }
47 }
```

```
----- CAT -----
Animal is eating
Animal is sleeping
Cat says: Meow Meow

----- DOG -----
Animal is eating
Animal is sleeping
Dog says: Woof Woof

=== Code Execution Successful ===
```